EXP. NO:4
01/08/26

PACKET SNIFFING USING RAW SOCKETS IN PYTHON

AIM:

To implement a packet sniffer using raw sockets in python that captures live network traffic and parses Ethernet, IP, and transport-layer (TCP/UDP, ICMP) headers.

STEPS:

STEP 1: Set up the lab environment

STEP 2: Understand the Packet structure you'll phrase

STEP 3: Create the Raw Socket

STEP 4: Parse the Ethernet Header

STEP 5: Parse the IP Header

STEP 6: Parse TCP/UDP/ICMP Headers

STEP 7: Write the Main Sniffing Loop

STEP 8: Run the Sniffer

STEP 9: Extend the lab

RESULT:

Thus, a packet sniffer using raw sockets that captures live network traffic and parses Ethernet, IP, and transport-layer headers implemented and achieved Successfully.

```
ge.py > ...
import socket
import struct
sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
host = socket.gethostbyname(socket.gethostname())
sniffer.bind((host, 0))
sniffer.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)
sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_ON)
print("Packet Sniffer Started...\n")
try:
    while True:
        raw_data, addr = sniffer.recvfrom(65535)
        ip_header = raw_data[0:20]
        iph = struct.unpack('!BBHHHBBH4s4s', ip_header)
        version = iph[0] >> 4
        ihl = iph[0] & 0xF
        ttl = iph[5]
        protocol = iph[6]
        source = socket.inet_ntoa(iph[8])
        destination = socket.inet_ntoa(iph[9])
        print("----------------------------------------")
        print("Version :", version)
        print("Header Length:", ihl * 4)
        print("TTL :", ttl)
        print("Protocol :", protocol)
        print("Source IP :", source)
        print("Destination :", destination)
        print("Packet Length:", len(raw_data))
except KeyboardInterrupt:
    sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_OFF)
    print("\nStopped")
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

Packet Sniffer Started...

----------------------------------------
Version : 4
Header Length: 20
TTL : 1
Protocol : 17
Source IP : 192.168.56.1
Destination : 224.0.0.251
Packet Length: 56
----------------------------------------
Version : 4
Header Length: 20
TTL : 1
Protocol : 17
Source IP : 192.168.56.1
Destination : 224.0.0.251
```